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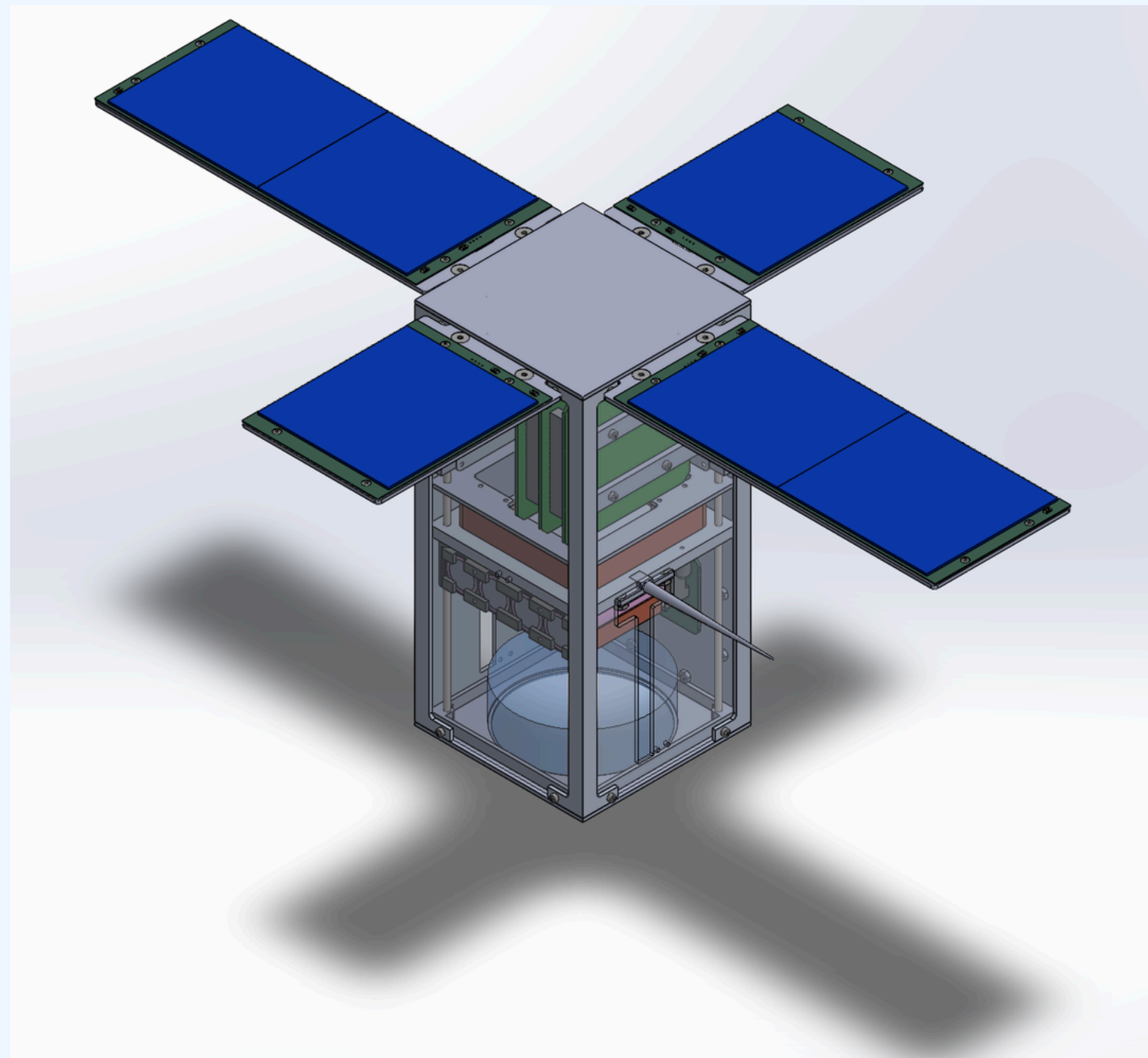


Figure 1. Full Satellite Assembly

Background

Mission Overview

CubeSat Structures is a team of aerospace & mechanical engineering students, responsible for the design and fabrication of the structural components of a 2U CubeSat, including: Bus frame, Solar Panels, ADCS (Magnetorquer Module & Reaction Wheels), Battery Pack, Monopole Antennas, Patch Antenna, Insulation, and Payload integration.

The SHAMROCK mission is to present a new application of an existing communication system for small sats, using Switch Beam Lens (SBL) Antennas

A SBL antenna uses a lens to direct radio waves in different directions (beam steering).

The SHAMROCK CubeSat's payload is a PCB with 10-12 antenna elements placed under an alumina GRIN lens.

Year in Review

Designed a new layout for subsystem placement within the 2U cube-sat, separate from previous models and prototypes.

Engineered a new aluminum bus frame structure through CAD modeling, followed by fabrication of a physical prototype to validate structural performance.

Developed a new solar panel deployment system using internal hinges, with panels designed by the IrishSat teams

Sewed two MLI models, one with Kapton and Mylar and another using just Mylar, to be used in thermal vacuum testing.

Designed an enhanced reaction wheel assembly model, achieving a thinner design compatible with the updated internal structure.

25/26 Projects

1-D Test: Test Bed

Beyond designing novel subsystems, it is important as a club that systems are validated using test beds. The purpose of this test bed is to determine how efficient the computing team's model is in controlling the satellite, which is done using a one-dimensional reaction wheel test.

The test bed is comprised of three layers that house the batteries, electronics, and reaction wheel from bottom to top. The test bed fits inside a hollowed bowling ball which sits on a frictionless stand to allow it to freely rotate and collect important rotational data.

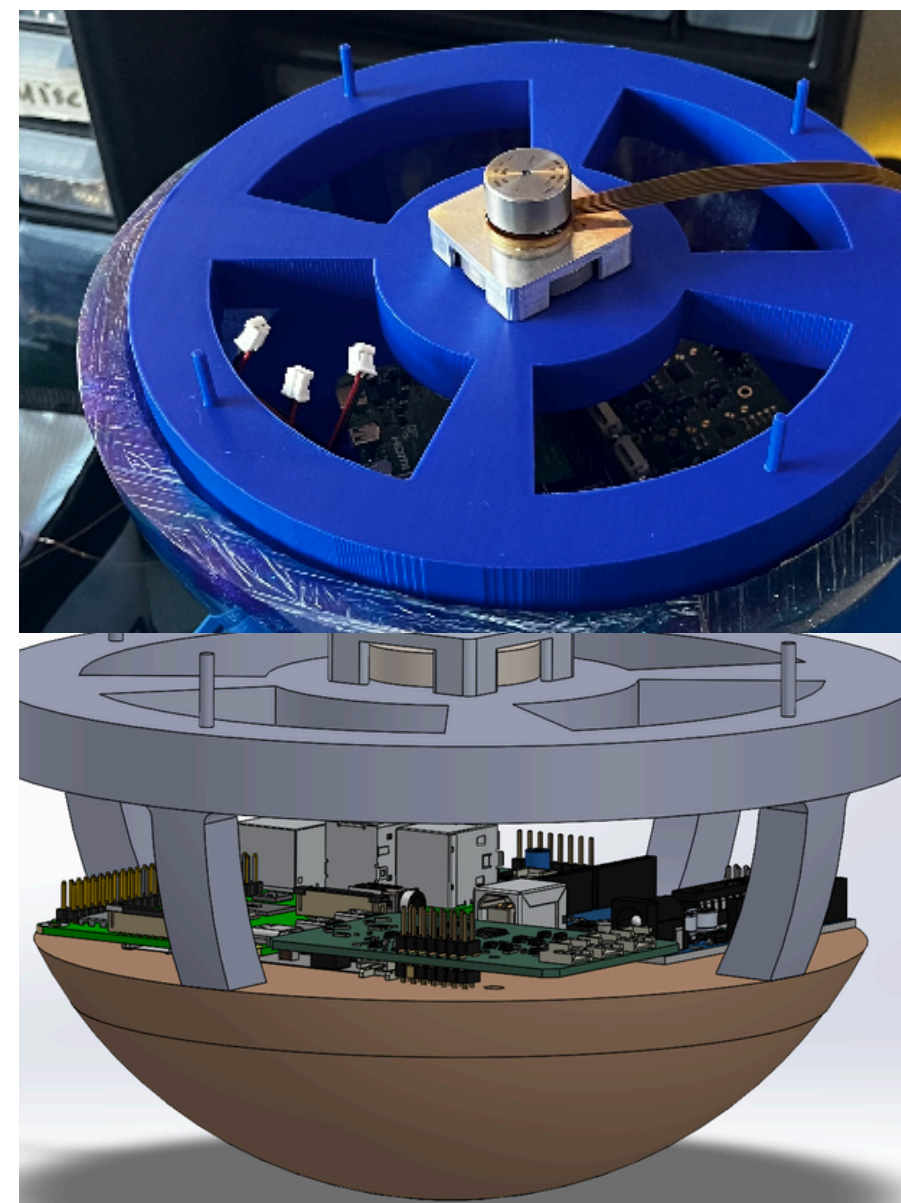


Figure 2. Test Bed and CAD

Insulation

Multi-Layer Insulation (MLI) minimizes heat transfer to the 2U structure in extreme environments. This 10-layer MLI is composed of layers of Mylar and polyester net spacers that avoid conduction between the layers. Two samples were created, one with Kapton and one without.

The MLI is sewed using fiberglass, a low-conduction, resistant thread that is able to withstand high temperatures. The MLI will be evaluated through regulated thermal testing in a vacuum chamber, where it is heated and tested for minimum thermal conduction.



Figure 3. MLI Samples

Bus Frame & Subsystem Placement

This year's goal was to improve subsystem placement and feasibility to reach internal components on the CubeSat.

A Modular Bus Frame was developed to achieve this. The frame involves 2 end caps, 4 walls, 4 rails, 4 internal rods, and adjustable shelves. The faces and rails attach to the end caps that are tapped, and the shelves supporting the internal components rest on the rods while supported with standoffs.

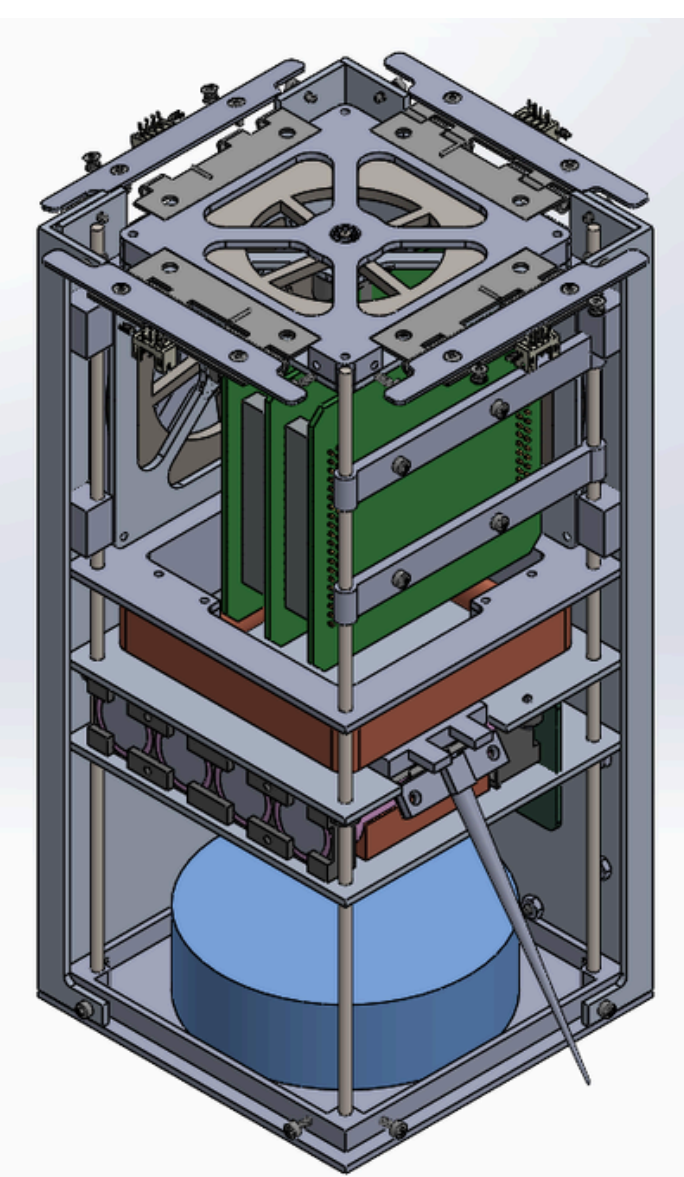
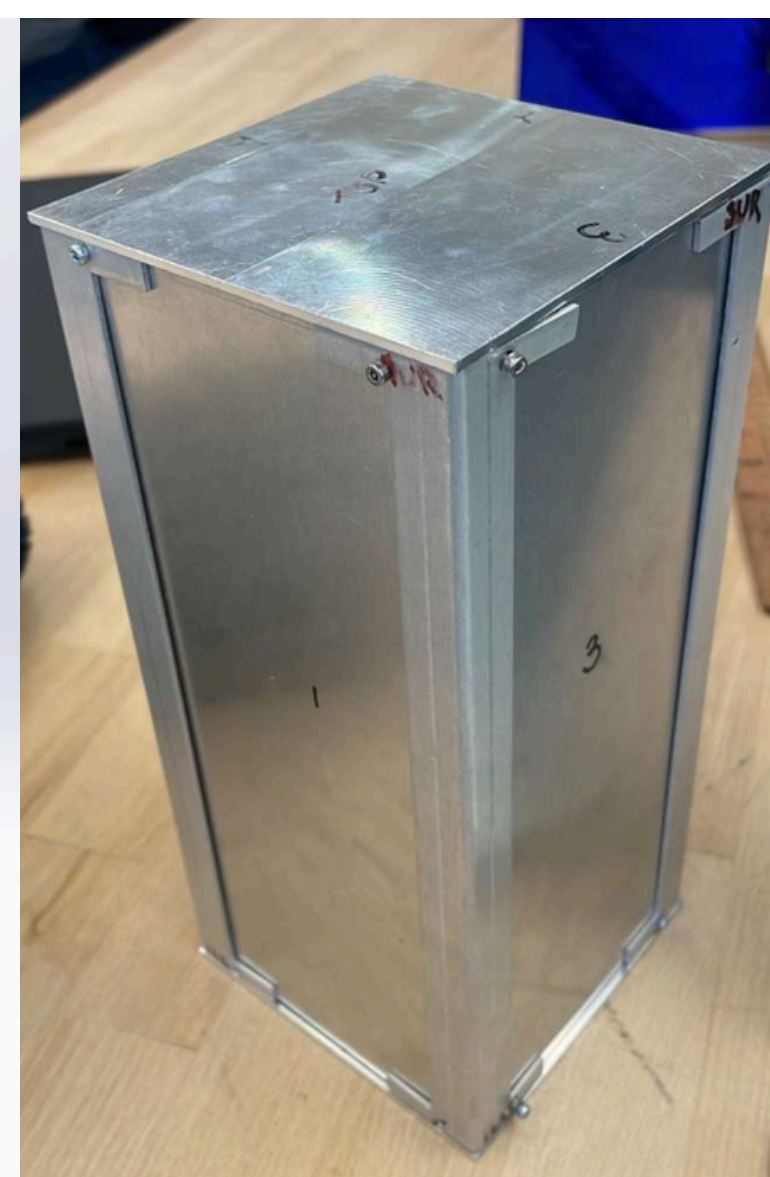
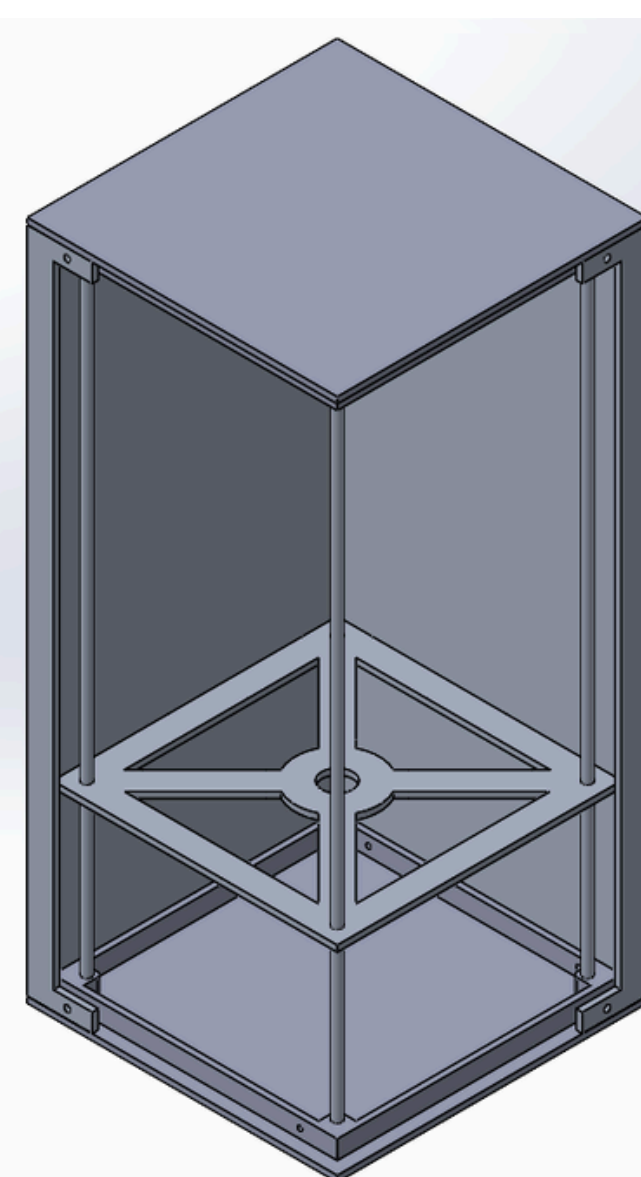


Figure 4. Bus Frame CAD, Metal Iteration, and Subsystem Placement

The internal components of the CubeSat are vertically stacked, avoiding attachment to the walls of the frame. The order of the stack goes as follows:

- Solar Panels
- Reaction Wheels
- PCB Stack
- Magnetorquer Module
- Monopole Antennas
- Battery Pack
- Patch Antenna (on wall)
- GRIN Lens

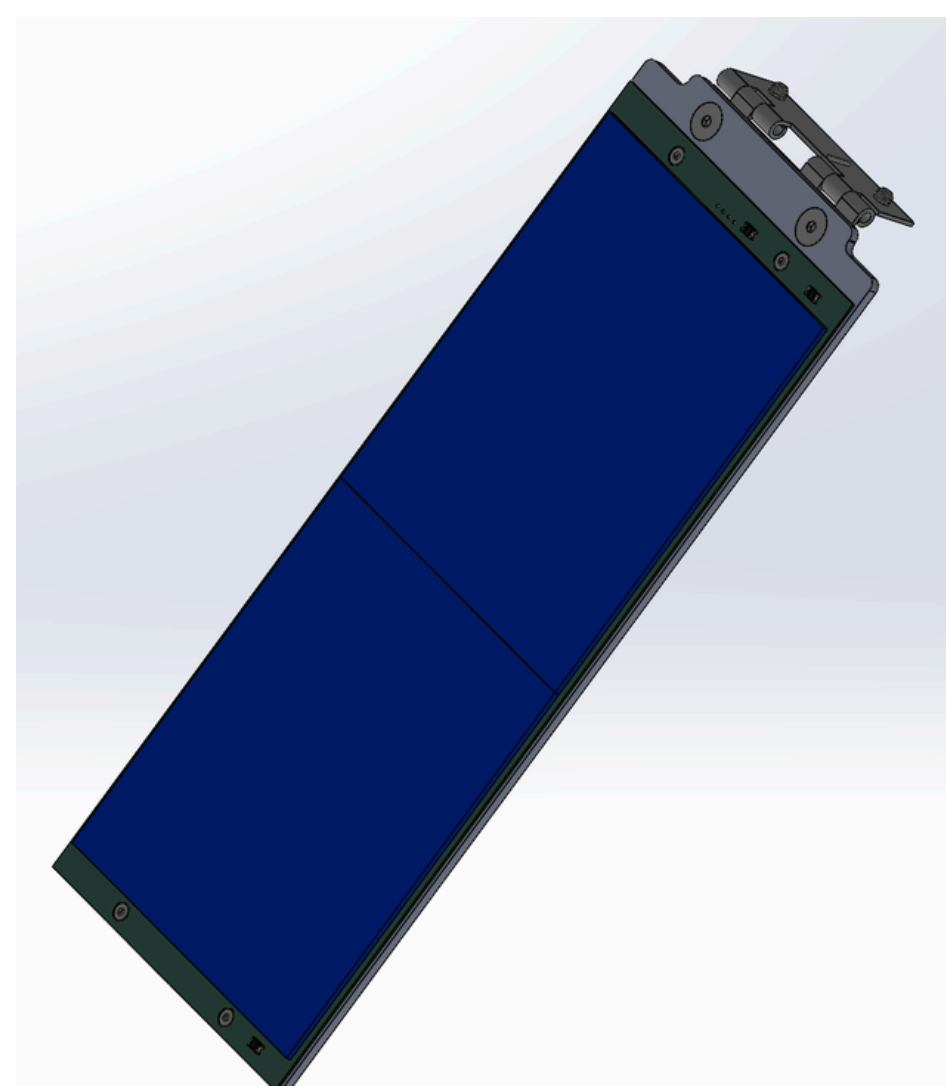


Figure 5. Solar Panel CAD

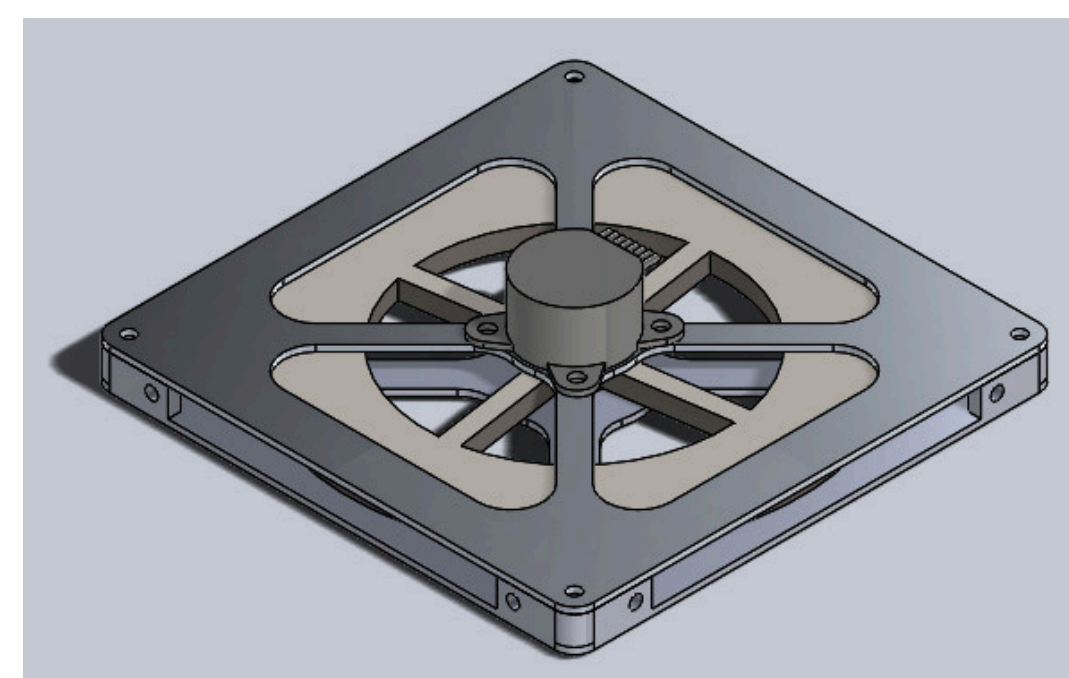
Solar Panel Deployment

Reliable solar panel deployment is critical to ensuring sufficient power for the CubeSat.

The four panels are held in the stowed configuration by a burnwire restraint and deploy via self-closing spring hinges. Passing current through the burnwire thermally severs it, releasing the panels. Stored hinge energy rotates the panels outward until stoppers define the final deployed angle.

This year's design mounts the hinges internally, reduces the backplate assembly thickness, and integrates the stoppers into the bus frame, resulting in a more compact CubeSat.

Reaction Wheel Module



The Reaction Wheel Module primarily provides momentum to the CubeSat to control attitude. It does so by spinning a spoked flywheel to high RPM. Three of these modules will be placed in each axis to provide three dimensional directional control.

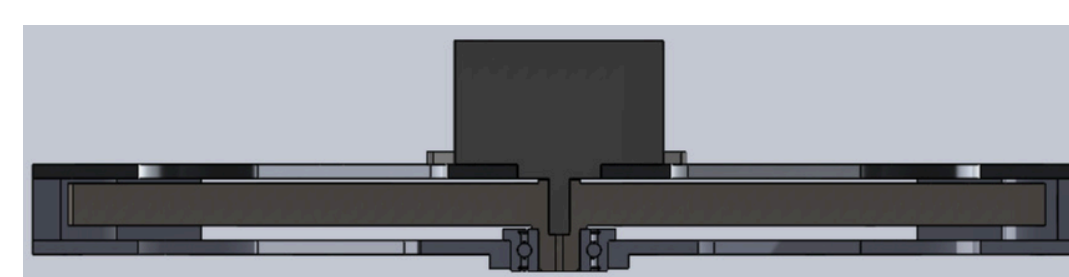


Figure 6. One Reaction Wheel CAD

The module is made of a thin aluminum cage with a steel flywheel inside spun by a motor and supported by a bearing on the opposite side. Three dimensional attachment points on each corner ease flexible assembly into the CubeSat.